Comprehensive Project Report: CIFAR-10 Image Classification with MobileNetV2

1. Project Title & Overview

This project implements a computer vision pipeline to classify images from the CIFAR-10 dataset into 10 distinct categories. It leverages Transfer Learning by fine-tuning a pre-trained deep neural network to achieve high accuracy efficiently.

2. Dataset Description

- **Source:** CIFAR-10 (automatically downloaded via TensorFlow/Keras).
- **Classes:** 10 categories (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck).
- **Format:** 32x32 color images.

3. Setup & Installation

Prerequisites

- Python 3.8+

Installation

Navigate to this directory and install the required dependencies:

```
pip install -r requirements.txt
```

4. Methodology & Pipeline

- **Data Preprocessing & Augmentation:**
  - Applied random horizontal flips, rotations, and zooms to artificially expand the training dataset and prevent overfitting.
  - Resized inputs dynamically to 96x96 to better fit the MobileNetV2 architecture.
- **Model Architecture (Transfer Learning):**
  - Loaded the pre-trained **MobileNetV2** (trained on ImageNet).
  - Froze the base model weights to retain its feature-extraction capabilities.
  - Added a custom classification head (GlobalAveragePooling, Dropout, and a Dense Softmax layer) optimized for our 10 classes.

5. Results & Evaluation

The model successfully learned to distinguish between the 10 classes.

- **Training Curves:** Validation accuracy and loss were tracked across epochs to ensure the model was learning without overfitting. *(Refer to `training_curves.png` in this folder for visual performance).*

6. Usage / How to Run

To train the model from scratch:

```
python train.py
```

To run inference on a new image (e.g., the provided OIP.jpg):

```
python inference.py --image_path OIP.jpg
```